Cross-Lingual Conflict Detection in Multilingual LLMs: Mechanistic Evidence for English-Anchored Representations

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Abstract

Multilingual large language models resolve entity-centric knowledge conflicts in ways that depend on linguistic affinity rather than resource abundance, yet the internal mechanisms underlying this behavior remain unexplored. We present a mechanistic investigation of cross-lingual conflict detection signals in the residual stream of three transformer architectures (Llama-3.1-8B, Qwen3-8B, Gemma-2-9B-IT) across two typologically distinct language pairs (EN↔ZH, EN↔JA). Linear probes trained on residual stream activations detect conflict signals at 70–77% balanced accuracy, with cross-lingual transfer exhibiting a consistent directional asymmetry: transfer toward English (X→EN) outperforms transfer away from English (EN→X) by 6.3–7.9 percentage points across all model–language-pair combinations tested. Sparse autoencoder (SAE) analysis reveals 18 cross-lingually shared conflict features concentrated in middle layers (Jaccard = 0.28) that diverge in late layers (Jaccard = 0.12, $p = 0.008$), while language-identity features occupy early layers with minimal overlap (4.3%). Representation steering using probe directions and SAE features causally shifts model outputs from parametric to contextual answers, with cross-lingual flip rates reproducing the same directional asymmetry. These converging lines of evidence—probing, feature decomposition, and causal intervention—support an English-anchored hypothesis: conflict detection is encoded in a language-universal mid-layer representation anchored to English, while language-specific resolution occurs during late-layer language transition.

1 Introduction

When multilingual LLMs encounter factual knowledge conflicts—cases where contextual information contradicts parametric knowledge—the language of presentation systematically influences which knowledge source prevails. Recent work by Zhao et al. (2026) (CLEAR) demonstrated that for entity-centric factual conflicts, linguistic affinity between the context and query languages determines conflict resolution outcomes, rather than the absolute resource abundance of either language. This finding raises a fundamental mechanistic question: *how are cross-lingual conflict signals encoded in the model's internal representations, and why does linguistic similarity play a decisive role?*

We investigate this question through three complementary analysis tracks applied to the residual stream of multilingual transformers. **Track 1** (probing) trains linear and MLP classifiers on residual stream activations to detect and transfer conflict signals across languages. **Track 2** (SAE decomposition) uses pre-trained sparse autoencoders to identify monosemantic conflict features and characterize their cross-lingual sharing patterns. **Track 3** (causal intervention) steers model behavior by manipulating conflict-related directions in activation space.

We formulate two competing hypotheses about the encoding of cross-lingual conflict signals:

- **H1 (English-anchored universal detection)**: Conflict detection signals are language-independent, encoded in a mid-layer shared representation space anchored to English. Late-layer language transition introduces language-specific interference that affects resolution but not detection. This predicts: (a) mid-layer cross-lingual feature sharing significantly exceeds late-layer sharing; (b) transfer toward English is consistently stronger than transfer away.
- **H2 (Language-pair-specific encoding)**: Conflict features are language-pair-specific from mid-layers onward, with linguistic distance directly encoded in conflict representations. This predicts: (a) no significant difference between mid-layer and late-layer feature sharing; (b) transfer patterns differ qualitatively across language pairs.

Our experiments on three architectures (Llama-3.1-8B, Qwen3-8B, Gemma-2-9B-IT) and two language pairs (EN↔ZH, EN↔JA) yield four principal findings:

1. Cross-lingual conflict signals are linearly detectable in the residual stream at 64–77% balanced accuracy across all directions and models, with permutation test $p < 0.001$.
2. Cross-lingual transfer is directionally asymmetric: X→EN exceeds EN→X by 6.3–7.9pp across all 4 model×language-pair combinations (3/4 statistically significant at 95% CI, 4/4 directionally consistent).
3. SAE conflict features are cross-lingually shared in middle layers (Jaccard = 0.28) but diverge in late layers (Jaccard = 0.12), with language-identity features occupying a distinct early-layer subspace (overlap rate 4.3%).
4. Representation steering causally shifts conflict resolution behavior, with cross-lingual flip rate asymmetry mirroring the probe transfer asymmetry.

These results collectively support H1: conflict detection is encoded in an English-anchored language-universal representation, with language-specific processing emerging during late-layer transition.

2 Related Work

Knowledge conflicts in LLMs. Knowledge conflicts arise when contextual information contradicts a model's parametric knowledge. Prior work has characterized monolingual conflict resolution behavior and identified internal signals predictive of which knowledge source prevails. However, the cross-lingual dimension remains largely unexplored. CLEAR (Zhao et al., 2026) established that for entity-centric conflicts, linguistic affinity—not resource abundance—determines resolution outcomes across 10 languages, but provided no mechanistic account.

Cross-lingual representations in multilingual models. Multilingual transformers develop shared cross-lingual representations in middle layers before diverging in late layers for language-specific decoding. Wendler et al. (2024) documented this "language transition" phenomenon across model families, finding that mid-layers form language-agnostic conceptual representations. Our work builds on this finding by examining how conflict-specific signals interact with this layer-wise structure.

Sparse autoencoders for mechanistic interpretability. Sparse autoencoders (SAEs) decompose neural activations into interpretable monosemantic features. Recent work has applied SAEs to identify knowledge-related features and demonstrated that manipulating these features can steer model behavior (SpARE). We extend this line to the cross-lingual setting, using pre-trained SAEs (Qwen-Scope) to decompose conflict signals into shared and language-specific components.

Representation steering. Activation steering modifies model behavior by adding direction vectors to the residual stream. While effective for monolingual settings, cross-lingual steering introduces additional challenges. We use both probe-derived directions and SAE feature directions for cross-lingual conflict steering, providing complementary causal evidence.

3 Method

3.1 Data Construction

We construct cross-lingual knowledge conflict datasets from ConflictQA, which contains entity-centric factual questions where contextual and parametric answers diverge. For each question, we obtain model responses in English (EN) and generate high-quality translations in Chinese (ZH) and Japanese (JA) using the target models themselves, followed by back-translation quality verification using COMET (wmt22-comet-da).

Table 1: COMET tercile ablation. Probe accuracy does not increase with translation quality, ruling out translation quality as a confound.

Model	Tercile	N	$\text{COMET}_{\text{mean}}$	Bal. Acc.	Δ vs Low
Llama-3.1-8B	Low	91	0.582	64.8%	—
	Med	91	0.714	66.1%	+1.3pp
	High	90	0.821	63.2%	−1.6pp
Qwen3-8B	Low	56	0.585	66.4%	—
	Med	56	0.700	67.7%	+1.3pp
	High	56	0.816	60.7%	−5.6pp

Translation quality validation. Back-translation COMET scores are computed for all translated samples. We perform tercile ablation analysis to verify that translation quality does not confound conflict detection: samples are split into low, medium, and high COMET terciles, and balanced accuracy is compared across groups. As shown in Table 1, both Llama-3.1-8B and Qwen3-8B show no positive correlation between COMET score and probe accuracy (Fisher $p > 0.69$), with the high-COMET tercile actually producing *lower* accuracy than the low-COMET tercile. This rules out translation quality as a confound.

3.2 Track 1: Residual Stream Probing

For each model, we extract residual stream activations at every layer for conflict and non-conflict samples in each language. We train logistic regression probes (L2-regularized) to classify conflict vs. non-conflict, using 5-fold stratified cross-validation with balanced accuracy as the metric. Cross-lingual transfer is evaluated by training on language A and testing on language B without fine-tuning. We assess significance using permutation tests ($n = 1000$) and bootstrap confidence intervals ($n = 100$) for the asymmetry gap.

MLP probes (single hidden layer, 256 units) are trained with identical splits to quantify non-linear components. Statistical significance is determined at $\alpha = 0.05$.

3.3 Track 2: SAE Feature Analysis

We apply pre-trained sparse autoencoders from Qwen-Scope to Qwen3-8B-Base activations. Conflict-relevant features are identified via mutual information between feature activations and conflict labels, yielding a ranked list from which the top 128 candidates are selected and filtered to retain features with robust conflict–non-conflict discrimination.

To disentangle conflict features from language-identity features, we apply LangFIR-style random-token filtering: features that activate consistently across random token sequences in a given language are classified as language-identity features, and their overlap with conflict features is quantified.

Cross-lingual feature sharing is measured by the Jaccard index of EN and ZH conflict feature sets, computed separately for early (0–33%), mid (33–67%), and late (67–100%) layer groups. The H1 vs. H2 prediction is tested by a paired permutation test comparing mid-layer and late-layer Jaccard.

3.4 Track 3: Causal Intervention via Steering

Probe-based steering. The weight vector of the best-layer linear probe defines a conflict direction in activation space. We add $\alpha \cdot \mathbf{v}$ to the residual stream at the optimal layer (where α is tuned to maximize flip rate while preserving coherence) and measure the rate at which model outputs shift from parametric to contextual answers.

SAE feature steering. We construct steering vectors from SAE conflict features (shared, EN-only, ZH-only, and combined sets) by clamping selected feature activations. Cross-lingual transfer is evaluated by applying EN-derived steering vectors to ZH inputs and vice versa.

Table 2: Cross-architecture probe results. EN→EN and ZH→ZH are within-language; EN→ZH and ZH→EN are cross-lingual transfer. Asym. gap = (ZH→EN) − (EN→ZH). All permutation test $p < 0.001$.

Model	EN→EN	ZH→ZH	EN→ZH	ZH→EN	Gap	95% CI	Sig.
Llama-3.1-8B	77.1%	71.2%	64.1%	71.3%	+7.2pp	[1.1, 13.6]	Yes
Qwen3-8B	75.3%	71.6%	65.0%	72.2%	+7.2pp	[−1.2, 13.4]	No (trend)
Gemma-2-9B-IT	76.4%	70.8%	64.7%	72.6%	+7.9pp	[2.3, 13.8]	Yes

Table 3: MLP gain in the weak transfer direction (EN→ZH). Models with better Chinese alignment show smaller non-linear gains.

Model	Linear	MLP	Gain	Interpretation
Llama-3.1-8B	64.1%	68.8%	+4.7pp	Strong non-linear component
Qwen3-8B	65.0%	67.3%	+2.3pp	Weak (better ZH alignment)
Gemma-2-9B-IT	64.7%	70.8%	+6.1pp	Moderate non-linear component

4 Experiments and Results

4.1 Cross-Architecture Conflict Detectability

We train linear probes on residual stream activations from three model architectures: Llama-3.1-8B, Qwen3-8B, and Gemma-2-9B-IT. Table 2 reports balanced accuracy for all four transfer directions (within-language and cross-lingual) along with the asymmetry gap and bootstrap confidence intervals.

All within-language probes achieve 70–77% balanced accuracy, substantially above the 50% chance baseline (permutation $p < 0.001$ for all). Cross-lingual transfer succeeds in all directions (64–73%), confirming that conflict signals are partially shared across languages.

The asymmetry gap is remarkably consistent: +7.2pp (Llama), +7.2pp (Qwen), and +7.9pp (Gemma), with a coefficient of variation below 6%. Two of three models reach statistical significance at 95% confidence; Qwen shows the same direction but the CI includes zero at $N \approx 1000$. We address this power issue in Section 4.6.

MLP non-linearity. MLP probes reveal direction-dependent non-linear components in the weak transfer direction (EN→ZH), as shown in Table 3. The gain is largest for Gemma (+6.1pp), moderate for Llama (+4.7pp), and smallest for Qwen (+2.3pp). This gradient anti-correlates with each model's Chinese pretraining data proportion, suggesting that stronger cross-lingual alignment reduces the non-linear component of cross-lingual conflict mapping.

4.2 Multilingual Extension: EN↔JA

To test whether the asymmetry generalizes beyond EN↔ZH, we replicate the full probe experiment on the EN↔JA language pair using Qwen3-8B and Llama-3.1-8B. Japanese is typologically distinct from Chinese (SOV word order, complex morphology, multiple writing systems) while belonging to the same CJK cultural sphere.

All four combinations in Table 4 show the same directional asymmetry: transfer toward English outperforms transfer away from English. This is consistent with H1 (English-anchored detection) and inconsistent with H2, which predicts language-pair-specific patterns. The JA asymmetry gap (6.3–6.7pp) is slightly smaller than the ZH gap (7.2pp), consistent with weaker Japanese representation in these models' training data. Three of four CIs exclude zero; the remaining one (Qwen EN↔ZH) is directionally consistent.

4.3 Layer-wise Representation Analysis

CKA and SVCCA analysis. We compute Centered Kernel Alignment (CKA) and Singular Vector Canonical Correlation Analysis (SVCCA) between EN and ZH activations at each layer to quantify cross-

Table 4: Multi-lingual asymmetry comparison. All four model×language-pair combinations show X→EN > EN→X.

Model	Pair	X→EN	EN→X	Gap	95% CI	Direction
Qwen3-8B	EN↔ZH	72.2%	65.0%	+7.2pp	[−1.2, 13.4]	ZH→EN wins
Qwen3-8B	EN↔JA	68.1%	61.8%	+6.3pp	[0.4, 12.5]	JA→EN wins
Llama-3.1-8B	EN↔ZH	71.3%	64.1%	+7.2pp	[1.1, 13.6]	ZH→EN wins
Llama-3.1-8B	EN↔JA	66.8%	60.1%	+6.7pp	[0.8, 13.1]	JA→EN wins

Table 5: CKA similarity between EN and ZH activations by layer group. All three models show high mid-layer similarity followed by a sharp late-layer drop.

Model	Early (0–33%)	Mid (33–67%)	Late (67–100%)	Mid–Late Drop
Llama-3.1-8B	0.71	0.83	0.52	−0.31
Qwen3-8B	0.68	0.81	0.56	−0.25
Gemma-2-9B-IT	0.70	0.82	0.53	−0.29
Mean	0.70	0.82	0.54	−0.28

lingual representation similarity. Table 5 reports mean similarity by layer group.

All three models exhibit the same pattern: EN–ZH representations are most similar in middle layers (CKA ≈ 0.82), with a sharp drop in late layers (CKA ≈ 0.54). The mean mid-layer SVCCA similarity is 0.78, dropping to 0.48 in late layers. This is consistent with the "language transition" phenomenon documented by Wendler et al. (2024).

The probe accuracy peak correlates negatively with the CKA transition point ($r = -0.71$), indicating that conflict signals are most detectable precisely where cross-lingual representations begin to diverge—the boundary between language-universal and language-specific processing. The language transition layer ratios are 0.75 (Llama), 0.60 (Qwen), and 0.68 (Gemma), suggesting model-specific transition dynamics.

PCA visualization of probe weights. PCA analysis of residual stream activations at the optimal probe layer reveals distinct conflict vs. non-conflict clusters. At mid-layers, EN and ZH conflict samples show high centroid overlap (cosine similarity = 0.78), supporting language-universal conflict encoding. At late layers, this overlap decreases substantially (cosine = 0.51), reflecting language-specific divergence. The silhouette score for conflict/non-conflict separation is higher for English (0.42) than Chinese (0.35), consistent with the English probe's higher within-language accuracy. PCA explains 38% of variance in two dimensions, indicating that conflict encoding is distributed but directionally coherent.

4.4 SAE Feature Analysis

Conflict feature identification. Using Qwen-Scope pre-trained SAEs on Qwen3-8B-Base, we identify conflict-relevant features via mutual information screening. From 65,536 total features, 128 candidates are selected, of which 47 pass robustness filtering. These features achieve AUROC = 0.81 for conflict vs. non-conflict discrimination and exhibit mean sparsity of 0.034.

Table 6 decomposes the identified features into EN-specific (20), ZH-specific (13), and cross-lingually shared (18) conflict features. The shared features achieve the highest AUROC (0.84), while language-identity features serve as a negative control with near-chance AUROC (0.52), confirming that conflict features carry semantic rather than language-identity information.

LangFIR deconfounding. LangFIR random-token filtering identifies 42 language-identity features, concentrated in early layers (68% in layers 0–33%). Conflict features, by contrast, concentrate in mid layers (61% in layers 33–67%). The overlap between conflict and language-identity features is minimal: only 2 features (overlap rate = 4.3%), and the layer distributions are significantly different ($p = 0.003$). Table 7 details the layer distribution.

Table 6: SAE conflict feature summary. Shared features achieve the highest AUROC; language-identity features (control) are near chance.

Category	Count	Sparsity	AUROC
EN-specific conflict	20	0.031	0.79
ZH-specific conflict	13	0.038	0.76
Cross-lingual shared	18	0.033	0.84
Language-identity (control)	42	0.025	0.52

Table 7: Feature type layer distribution. Language-identity features concentrate in early layers; conflict features concentrate in mid layers with minimal overlap.

Feature Type	Early (0–33%)	Mid (33–67%)	Late (67–100%)	Total
Language-identity	29 (69%)	9 (21%)	4 (10%)	42
Conflict (shared)	2 (11%)	11 (61%)	5 (28%)	18
Conflict (EN-only)	1 (5%)	10 (50%)	9 (45%)	20
Conflict (ZH-only)	1 (8%)	7 (54%)	5 (38%)	13

Cross-lingual Jaccard analysis. The Jaccard index of EN and ZH conflict features varies significantly across layer groups (Table 8). Mid-layer Jaccard (0.28) substantially exceeds late-layer Jaccard (0.12), with a difference of 0.16 that is statistically significant (paired permutation test $p = 0.008$). This directly supports H1 over H2: conflict features are shared across languages in mid layers but diverge after the language transition in late layers.

4.5 Causal Intervention: Representation Steering

Probe-based steering. Using the linear probe's weight vector as a steering direction, we add a scaled perturbation to the residual stream at the optimal layer and measure the rate at which model outputs flip from parametric to contextual answers (or vice versa). Table 9 reports flip rates across models and directions.

All steering conditions produce flip rates 3.4–6.5× above random baselines, establishing a causal link between the detected conflict signal and conflict resolution behavior. Monolingual EN steering achieves the highest flip rate (31.2%), followed by monolingual ZH (26.8%), cross-lingual ZH→EN (24.3%), and cross-lingual EN→ZH (17.8%). The cross-lingual asymmetry gap (Llama: 24.3% − 17.8% = 6.5pp) mirrors the probe transfer asymmetry, providing causal evidence that the asymmetry reflects the structure of conflict encoding, not merely a statistical artifact of probe training. Coherence is preserved at 91%.

SAE feature steering. SAE-based steering enables more targeted intervention at the feature level. Table 10 compares monolingual and cross-lingual steering effectiveness using SAE conflict features.

Cross-lingual SAE steering achieves a transfer efficiency of 69% (cross-lingual flip rate / monolingual flip rate), indicating that English conflict features encode substantial cross-lingual information. The directional asymmetry persists: ZH→EN steering (31.0%) outperforms EN→ZH steering (24.0%) by 7.0pp, consistent with probe-based results. Random feature steering serves as a negative control at 5.2%.

Feature-set ablation. To further test H1, we compare steering effectiveness when using only shared features, only language-specific features, and the full combined set (Table 11).

Shared features alone achieve 28.0% cross-lingual flip rate, while language-specific features yield only 9–11%. The combined set provides a marginal 18% lift over shared-only, indicating that cross-lingual conflict steering is primarily driven by the language-universal feature subspace. This is the most fine-grained causal evidence for H1: the shared features identified in Section 4.4 are not merely correlated with conflict signals but causally drive cross-lingual conflict resolution.

Table 8: Layer-group Jaccard analysis. Mid-layer sharing significantly exceeds late-layer sharing ($p = 0.008$), supporting H1.

Layer Group	EN feat.	ZH feat.	Shared	Jaccard	95% CI
Early (0–33%)	3	3	1	0.20	[0.04, 0.41]
Mid (33–67%)	21	18	8	0.28	[0.16, 0.42]
Late (67–100%)	14	10	3	0.12	[0.03, 0.24]

Table 9: Probe-based steering flip rates. Cross-lingual steering reproduces the asymmetry: ZH→EN > EN→ZH. All conditions substantially exceed the random baseline (≈5%).

Model	Direction	Flip Rate	Random Baseline	Lift
Llama	EN (mono)	31.2%	4.8%	6.5×
Llama	ZH (mono)	26.8%	5.1%	5.3×
Llama	ZH→EN (cross)	24.3%	4.9%	5.0×
Llama	EN→ZH (cross)	17.8%	5.2%	3.4×
Qwen	EN (mono)	28.9%	4.6%	6.3×
Qwen	ZH→EN (cross)	22.1%	4.7%	4.7×
Qwen	EN→ZH (cross)	19.2%	5.0%	3.8×

4.6 Robustness and Scaling

Larger dataset validation. Expanding ConflictQA sampling to ∼2,200 samples per model narrows confidence intervals by ∼28% (Table 12). Critically, Qwen's CI shifts from [−1.2, 13.4] to [1.8, 12.3], achieving statistical significance. Point estimates remain stable (7.0pp vs. 7.2pp), confirming that the effect size is robust and Qwen's initial non-significance was a power issue.

Label noise robustness. We systematically inject 5–20% label noise into the training set and measure probe accuracy degradation (Table 13). The probe degrades gracefully at ∼2.6pp per 5% noise, maintaining >73% accuracy at 10% noise. Given that our verification pipeline's estimated label error rate is <5%, this provides ample safety margin. Cross-lingual noise sensitivity is similar across directions (1.3× for EN→ZH, 1.1× for ZH→EN), indicating that robustness is an intrinsic property of the probe rather than language-dependent.

5 Discussion

H1 vs. H2 resolution. Three independent lines of evidence converge on H1 (English-anchored universal detection):

1. **Probe transfer asymmetry**: All 4 model×language-pair combinations show X→EN > EN→X (6.3–7.9pp), consistent with English as the representation anchor.
2. **SAE Jaccard analysis**: Mid-layer conflict features are significantly more shared across languages (Jaccard = 0.28) than late-layer features (Jaccard = 0.12, $p = 0.008$), consistent with language-universal mid-layer encoding diverging during late-layer transition.
3. **Steering asymmetry**: Causal intervention via both probes and SAE features reproduces the directional asymmetry, ruling out statistical artifacts and establishing that the asymmetry reflects the causal structure of conflict encoding.

H2 is not supported: the asymmetry direction is identical for EN↔ZH and EN↔JA (two typologically distinct pairs), and mid-layer feature sharing significantly exceeds late-layer sharing.

Table 10: SAE feature steering: cross-lingual transfer. EN features transferred to ZH retain 69% of monolingual effectiveness. The ZH→EN direction again outperforms EN→ZH.

Condition	EM_C	EM_M	Flip Rate	Coherence
EN mono (baseline)	0.42	0.31	35.0%	0.93
EN→ZH (cross)	0.29	0.22	24.0%	0.87
ZH→EN (cross)	0.34	0.26	31.0%	0.89
Random features (control)	0.05	0.04	5.2%	0.94

Table 11: Feature-set steering ablation. Shared features drive the majority of cross-lingual steering; language-specific features contribute marginally.

Feature Set	Target Lang	Flip Rate	vs. Shared
Shared (18 features)	ZH	28.0%	1.0×
EN-only (20 features)	ZH	11.0%	0.39×
ZH-only (13 features)	EN	9.0%	0.32×
Combined (all 51)	ZH	33.0%	1.18×

Connection to CLEAR. Our findings provide a mechanistic account for CLEAR's observation that linguistic affinity determines entity-centric conflict resolution. The English-anchored representation structure means that languages with higher similarity to English have more overlapping conflict features, enabling more effective cross-lingual conflict transfer. The mid-layer language-universal encoding explains why conflict *detection* is robust across languages, while the late-layer language-specific transition explains why conflict *resolution* depends on linguistic similarity.

Preliminary typological correlation. With only two language pairs (EN↔ZH, EN↔JA), we observe that steering effectiveness is directionally consistent with linguistic similarity (CLEAR similarity: EN–ZH 0.72, EN–JA 0.58; steering flip rate: 28.0%, 23.0%). This is suggestive but insufficient for statistical claims; validation with additional language pairs spanning diverse typological families is needed.

Limitations. (1) Our analysis covers three architectures from the same dense transformer family; MoE architectures may exhibit different patterns. (2) The two language pairs tested are both Asian languages; European languages would strengthen typological diversity. (3) SAE analysis uses pre-trained SAEs on a base model, while probing uses instruct-tuned models; the cross-validation between tracks partially mitigates but does not eliminate this mismatch. (4) Steering flip rates of 17–35% indicate that the linear/sparse approximation captures only part of the conflict mechanism; non-linear components remain unaddressed by our intervention framework.

6 Conclusion

We present the first mechanistic investigation of cross-lingual knowledge conflict signals in multilingual LLMs. Through residual stream probing across three architectures and two language pairs, SAE feature decomposition, and causal representation steering, we establish that: (1) conflict signals are linearly detectable in the residual stream with consistent cross-lingual transfer; (2) transfer is directionally asymmetric, favoring English as the anchor language; (3) conflict features are shared across languages in mid layers but diverge in late layers; and (4) manipulating these features causally shifts conflict resolution behavior. These findings support an English-anchored hypothesis for conflict encoding and provide mechanistic grounding for the role of linguistic affinity in cross-lingual conflict resolution.

Table 12: Sample size scaling. Doubling the dataset narrows CIs by ∼28% and brings Qwen to significance.

Model	N	Asym. gap	95% CI	Sig.?
Llama	∼1,000	+7.2pp	[1.1, 13.6]	Yes
Llama	∼2,200	+7.4pp	[3.1, 11.8]	Yes
Qwen	∼1,000	+7.2pp	[−1.2, 13.4]	No
Qwen	∼2,200	+7.0pp	[1.8, 12.3]	Yes

Table 13: Label noise robustness (Llama EN→EN). Graceful degradation at ∼2.6pp per 5% noise.

Noise Level	Bal. Acc.	Δ vs. 0%	Rel. Drop
0%	77.1%	—	—
5%	75.4%	−1.7pp	2.2%
10%	73.1%	−4.0pp	5.2%
15%	70.2%	−6.9pp	8.9%
20%	66.8%	−10.3pp	13.4%

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